

European Girls Math Olympiad 2023 — P3/6

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Let k be a positive integer. Lexi has a dictionary $\mathcal{D}$ consisting of some k -letter strings containing only the letters A and B. Lexi would like to write either the letter A or the letter B in each cell of a $k \times k$ grid so that each column contains a string from $\mathcal{D}$ when read from top-to-bottom and each row contains a string from $\mathcal{D}$ when read from left-to-right.

What is the smallest integer m such that if $\mathcal{D}$ contains at least m different strings, then Lexi can fill her grid in this manner, no matter what strings are in $\mathcal{D}$?

Solution

The answer is that the smallest such m is 2^{k-1} .

Let s be a string comprised of the letters A and B. Define the *opposite* of s to be a string $\bar{s}$ which is the same string, but just with any A's replaced by B's and vice versa.

Claim 1: If $\mathcal{D}$ contains a string and it's opposite, then Lexi can fill in the grid.

Proof: We have both $s, \bar{s} \in \mathcal{D}$. WLOG let s start with an A. We will show that Lexi can fill in the grid by construction.

Let (i, j) be the cell in the i -th row (from the top) and the j -th column (from the left). Let $L(i, j)$ be the letter at cell (i, j) . Lexi can start by filling in the top row using s (from left to right), and by filling in the leftmost column using s (from top to bottom). Then for each column, she can fill in the column (from top to bottom) using s if the topmost cell of that column is A, and using $\bar{s}$ otherwise.

Let $2 \leq u, v \leq k$ be integers. Now consider a cell (u, v) .

Case 1: $L(1, v) = L(u, 1)$. Suppose WLOG that $L(1, v) = L(u, 1) = A$. Now we want the u -th row to be s (from left to right), and we want the v -th column to be s (from top to bottom). Thus for the cell (u, v) to be valid, we must have $s_v = s_u$, which is true since it is implied from $L(1, v) = L(u, 1)$.

Case 2: $L(1, v) \neq L(u, 1)$. Suppose WLOG that $L(1, v) = A$ and $L(u, 1) = B$. We want the u -th row to be $\bar{s}$ (from left to right) and we want the v -th column to be s (from top to bottom). Thus for the cell (u, v) to be valid we must have $\bar{s}_v = s_u$, which is true since $L(1, v) \neq L(u, 1) \implies s_v \neq s_u \implies \bar{s}_v = s_u$. $\square$

Notice that out of the 2^k possible distinct strings exactly 2^{k-1} start with A, and exactly 2^{k-1} start with B. Now we cannot have $m < 2^{k-1}$ because otherwise, we can just take $\mathcal{D}$ to be m different strings starting with A other than $AAA \dots A$. This fails because after filling in the leftmost column with a string, one of the rows must start with the letter B, but no such string exists in $\mathcal{D}$, contradiction.

Note that by definition every string has a unique opposite. There are 2^{k-1} such unordered pairs of strings and their opposites $(s, \bar{s})$. However if $m = 2^{k-1}$ then, either we have picked a string of the form $AAA \dots A$, $BBB \dots B$ or we have picked a pair of opposite strings. If the latter case is true then Lexi can fill the grid due

to claim 1. If the former case is true then Lexi can trivially fill in the entire grid with only one letter, A if $AAA \dots A \in \mathcal{D}$ and B otherwise, this can be easily checked to satisfy the necessary requirements for a valid grid. ■

Comments: Finding the actual solution isn't all that difficult, the real challenge is in properly describing one's construction as I did in claim 1.